



# Kulpio

**Waste less food. Save money. In any language.**

A privacy-first Progressive Web App that tracks your fridge, tells you what expires first, and helps you cook it before it's wasted — offline, on any phone, in 33 languages.

**FIRSTSTEP × INFOMATRIX 2026**

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## WHY I BUILT THIS

# It started in my own fridge



I kept watching good food go bad — bought with the best intentions, pushed to the back of the shelf, forgotten, then thrown out.

That bothered me twice over: it's money in the bin, and wasted food is a real climate problem. **So I built the tool I wanted** — one that just tells me what to use next, before it's too late.

The idea in one line

**“Don't let me forget the food I already own.”**

## THE PROBLEM

# A third of our food is wasted — mostly at home

~1/3

of all food produced is lost or wasted (FAO)

#1

households are among the largest single contributors

Forget

the main cause at home: bought, forgotten, expired



It's an information & attention problem — not a preservation problem.  
That's exactly what an app can fix.

# Built for the Sustainable Development Goals



PRIMARY

## Responsible Consumption & Production

Kulpio's core mission — cut the food households throw away.



## Zero Hunger

Use more of the food we buy; value it instead of binning it.



## Climate Action

Less wasted food means fewer needless food-related emissions.



**UN Target 12.3:** halve per-capita global food waste by 2030. Kulpio goes straight at household waste — the largest, most-forgettable slice.

## THE IDEA

# Track it, know what's going off, cook it first

01



### Add in seconds

Scan a barcode, read a label or receipt with AI, or paste a list — no typing marathon.

02



### Know what expires

Printed dates beat estimates. Live freshness colours and a 'week-ahead' strip.

03

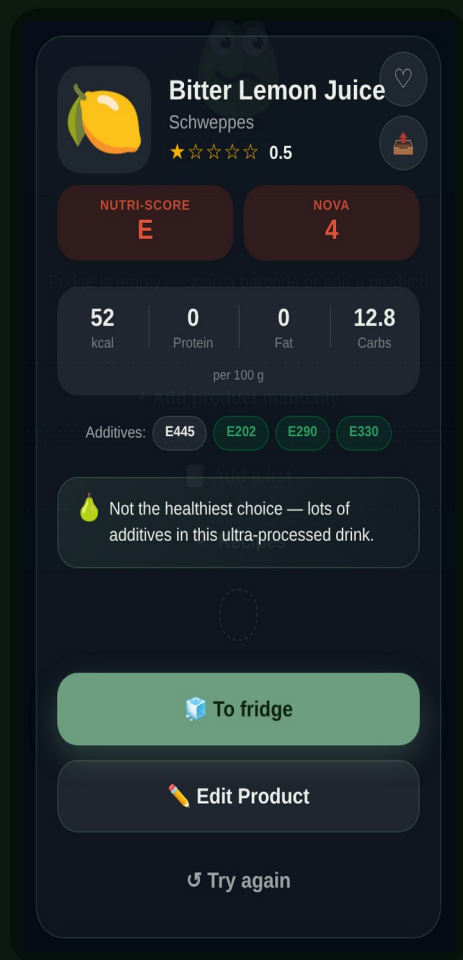


### Use it in time

Cook what's going off first; mark Used / Wasted / Froze and watch the money you save.

SEE IT

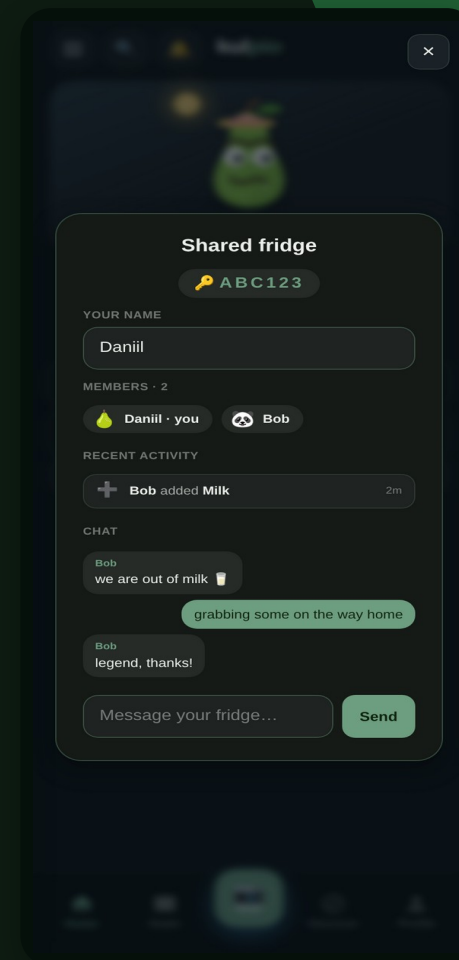
# One app, offline, on any phone



**Scan → product card**  
additives, Nutri-Score, verdict



**Your impact**  
waste cut & money saved



**Shared fridge**  
members, activity & chat

## FEATURES

# What Kulpio does



### Effortless input

Barcode, AI label & receipt scan, or paste a list



### Storage sections

Fridge, freezer & pantry with live freshness



### Recipes

Cook from your ingredients, with nutrition estimates



### Shared fridge

One code links a household + activity feed & chat



### Motivation

Achievements, XP levels, monthly recap & impact



### 33 languages

Full RTL support; installs to the home screen

## METHODOLOGY

# How Kulpio decides when food expires

*Printed dates are ground truth — Kulpio only estimates when there's no date to read.*

1

### Printed date wins

Read from the pack (OCR + model) or typed by you. Always takes priority.

2

### Offline shelf-life table

Built-in per-food estimate, adjusted for storage (freezer + ~90 d; opened shortens).

3

### AI refine

When online, the model refines the estimate from the product name.

### AI degrades gracefully

Anthropic Claude (best)



Cloudflare Workers AI — Llama (free)



Clean JSON error — never blocks



# Built as one offline file + one worker



## Client

- **Single static HTML** — ~1.3 MB — markup, CSS, JS, 33 langs
- **No build step** — no framework, no bundler
- **Service worker** — precache → fully offline
- **Web Crypto** — hashing & secure tokens



## Backend

- **Cloudflare Worker** — serves the app AND the API
- **D1 (SQLite)** — accounts, sync, households, community
- **Web Push (VAPID)** — daily expiry reminders
- **Rate-limited auth** — PBKDF2 · 5 tries / 15 min

**Deploy:** one Cloudflare Worker serves static assets + API on a single origin — installs from static hosting, runs offline.

## VS. EXISTING APPS

# Why Kulpio is different

|                             | Offline, no install | AI label/receipt | Shared fridge + chat | 33 languages | Privacy-first |
|-----------------------------|---------------------|------------------|----------------------|--------------|---------------|
| Kulpio                      | ✓                   | ✓                | ✓                    | ✓            | ✓             |
| Inventory apps (NoWaste...) | ✗                   | partial          | ✗                    | few          | varies        |
| Too Good To Go              | ✗                   | ✗                | ✗                    | some         | account       |
| Whisk / Samsung Food        | ✗                   | partial          | ✗                    | some         | cloud-first   |

**The combination** — *offline single-file PWA + AI input + a social shared fridge + privacy* — is the edge.

# Private by design, hardened by default



## Local-first

Your fridge stays in the browser — nothing leaves unless you choose a networked feature.



## GDPR export

One tap downloads everything the app holds about you as JSON.



## Real deletion

Delete account wipes the server account, sessions & synced data.



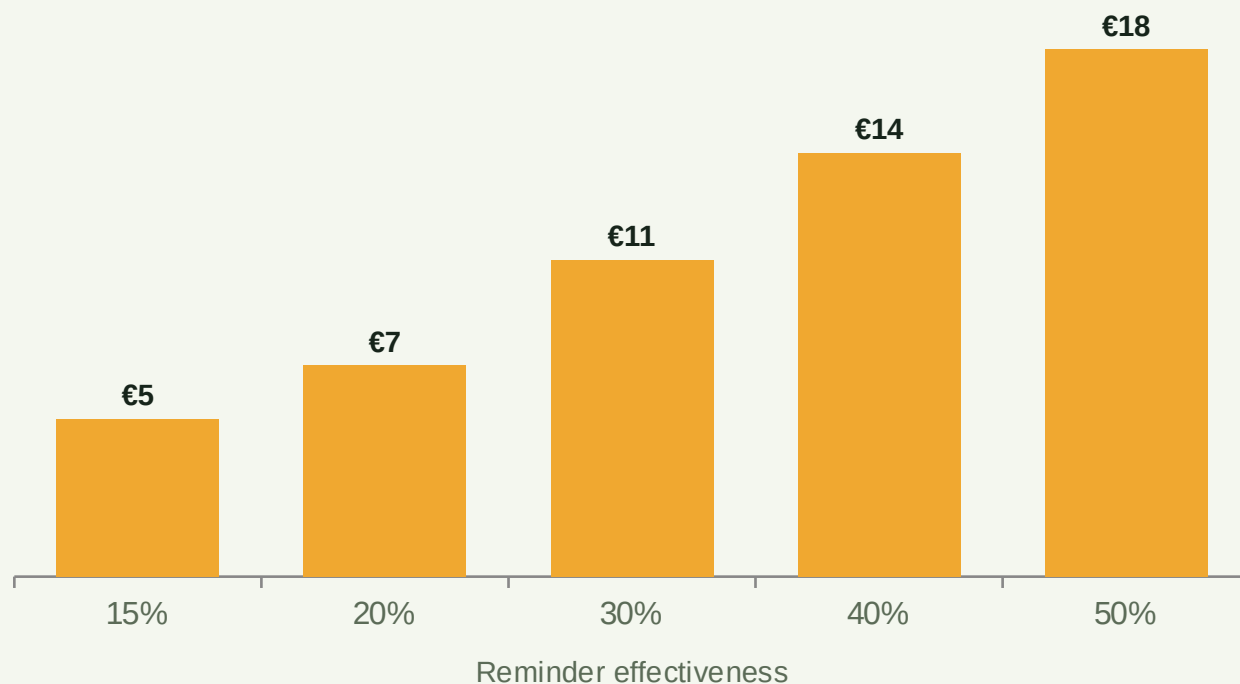
## Hardened auth

PBKDF2 hashing, opaque tokens, and login rate-limiting (5 / 15 min).

## IMPACT

# Projected savings — and how we'll measure it

PROJECTION · MODEL, NOT A STUDY



**At a mid-range 30% reminder effectiveness:**

≈ **30%** fewer wasted items  
≈ **€10-11** saved / household / month  
≈ **€128** / household / year

**Measured for real by a pilot:** a within-subject micro-study (baseline vs. Kulpio week, N = 3-15) using the app's built-in CSV export.

# How it evolved — and what I had to solve

01

## Offline core

Single-file PWA + shelf-life table.

### Challenge solved

Had to run with no network → service-worker precache.

02

## Trustworthy dates

AI label reading + estimates.

### Challenge solved

Estimates felt unreliable → printed dates always win.

03

## Kill the friction

Barcode, receipt scan, paste-a-list.

### Challenge solved

Typing was tedious → four fast input paths.

04

## Social & safe

Shared fridge, accounts, GDPR, 33 langs.

### Challenge solved

Two phones clobbered data → server-side merge.

# Where it goes — and how it stays alive



## Open & sustainable

**Free under GNU AGPL v3** for open use.

**Paid commercial licence** for closed-source use —  
per product:

**Indie €250 · Business €1,500 · Enterprise €5,000+**

*Keeps the project open while funding its future.*



## Roadmap

- Run the real user pilot & publish results
- Validate the offline shelf-life table
- Email verification (add an email provider)
- Field-level merge for multi-device edits
- Shareable monthly impact report



# Waste less. Save more. Together.

Kulpio — a food-freshness tracker that fits in your pocket, in your language.

SDG 12

SDG 2

SDG 13